inClass-L6

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CV2090
Probability & Statistics for Civil Engineers
(Jul – Nov 2026)

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L6
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Flow

- Module-1 ends today
  - Bayes' rule
  - Total probability rule versus Bayes' rule

Criminal Court
→ Beyond a reasonable doubt

Civil Court
→ Preponderance of the evidence
(more likely true than not)

Detainment → Reasonable suspicion

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Module I: Chapter 1 (M1C1)

Module 1: **Probability** – Understanding Uncertainty

Foundations of Probability ⇨ RVs & Probability Models ⇨ Multivariate Probability ⇨ Modeling Extremes

Foundations of Probability:
- Sample spaces and events
- Set algebra, Venn diagrams, De Morgan's
- Axioms of probability
- Probability rules (complements, unions, inclusion-exclusion)
- Conditional probability, independence
- Total probability theorem
- Bayes' theorem

RVs & Probability Models:
- Random variables
- Discrete vs continuous
- PMF/PDF/CDF; Support
- Common dist.: Uniform/Exp.
- Expectation, variance
- Higher moments

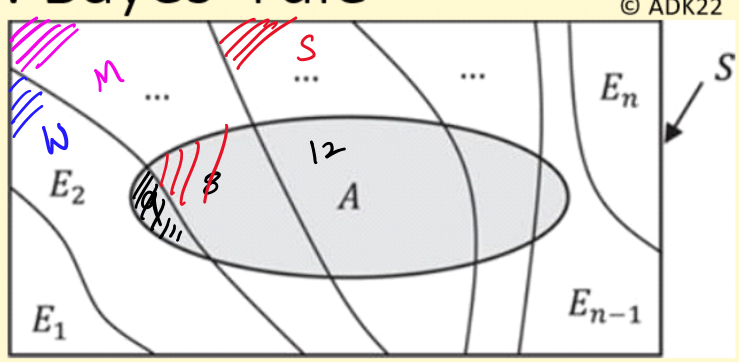
Multivariate Probability:
- Joint distributions, marginals
- Conditional distributions
- Covariance, correlation, independence
- Functions of rv, Jacobian
- uncertainty prop
- Monte Carlo (brief)
- FOSM

Modeling Extremes:
- CLT. Normal, lognormal
- Additive, multiplicative processes
- Bernoulli process, binomial, poisson
- Arrival processes
- Extreme value theory, block maxima
- Gumbel, return periods,
- Heavy tails
- Non-stationarity, climate extremes

Probability rules: Bayes' rule

- If $E_1, E_2, \ldots, E_n$ are n MECE events,

Partitioning



$$Pr(AE_i) = Pr(E_i|A) \cdot Pr(A) = Pr(A|E_i) \cdot Pr(E_i)$$

$$\Rightarrow Pr(E_i|A) = \frac{Pr(A|E_i) \cdot Pr(E_i)}{Pr(A)}$$

$$= \frac{Pr(A|E_i) \cdot Pr(E_i)}{\sum_{j=1}^{n} Pr(A|E_j)\, Pr(E_j)}$$

Probability rules: Bayes' rule

- Assume in the previous example that the building has sustained damage. What can we learn about the intensity of the earthquake?

$Pr(W) = 90\%$ $\quad Pr(M) = 8\%$ $\quad Pr(S) = 2\%$

$Pr(D|W) = 1\%$ $\quad Pr(D|M) = 10\%$ $\quad Pr(D|S) = 60\%$

$Pr(D) = 90\% \times 1\% + 8\% \times 10\% + 2\% \times 60\%$

$= 0.9\% + 0.8\% + 1.2\% = 2.9\%$

$Pr(W|D) = 0.9/2.9 = 31\% \leftarrow 90\%$

$Pr(M|D) = 28\% \leftarrow 8\%$

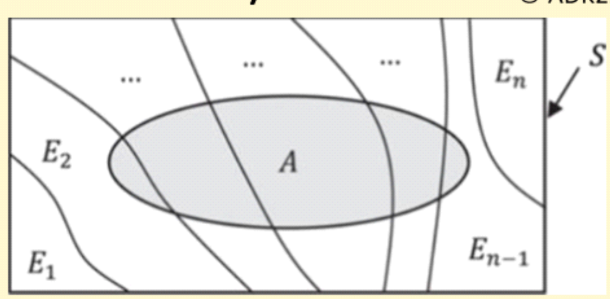
$Pr(S|D) = 41\% \leftarrow 2\%$

Posterior

Prior

Total Probability Rules & Bayes' rule

$E_1, E_2, \ldots, E_n$ are (MECE) partitions of S.



Total Probability Rule
- Forward Prediction
- What will happen?

Baye's Rule
- Backward Prediction / Inference
- Now that this event has occurred, what caused it?

Total Probability and Bayes' Rule irl

Total Probability Rule

- Pr(Cancer) $\longrightarrow$ Insurance Premium
- Pr(Cancelled flight) $\longrightarrow$ Ticket pricing
- Pr(Building collapse) $\longrightarrow$ Building design standards

Bayes' Rule

- Now that this patient has a symptom
  - What is Pr(cancer)?
    - operate / Prescribe
    - More Tests?

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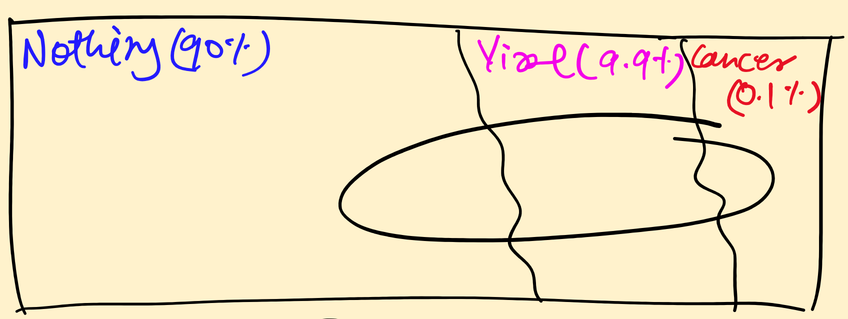
Bayes' Rule: example



Pr(B·Ache | Nothing) = 5%

Pr(B·Ache | Viral) = 50%

Pr(B·Ache | Cancer) = 90%

Calculate Pr(B·Ache)

$Pr(BA) = Pr(BA|N) \times Pr(N) + Pr(BA|V) \times Pr(V) + Pr(BA|C) \times Pr(C)$

$= 4.5\%$ 4.95% 0.09%

$= 9.54\%$

$Pr(N | B\cdot Ache) = 47\%$ $Pr(V|BA) = 52\%$ $Pr(C|BA) = 1\%$

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Murder trial of O. J. Simpson

O. J. Simpson

57 languages

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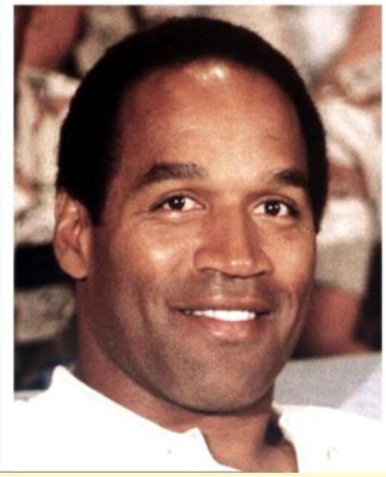
"The Juice" redirects here. For other uses, see Juice (disambiguation).

O. J. Simpson



Orenthal James Simpson (July 9, 1947 – April 10, 2024), nicknamed **"the Juice"**,[b] was an American football player, actor, and media personality. He played in the National Football League (NFL) for 11 seasons—nine with the Buffalo Bills—and is regarded as one of the greatest running backs of all time. His success is widely considered to be overshadowed by his two criminal charges for the 1994 murders of his ex-wife Nicole Brown and her friend Ron Goldman, and the contentious criminal trial in which he was acquitted on both counts.

Simpson played college football for the USC Trojans, winning the 1968 Heisman Trophy as a senior. He was selected first overall by the Bills in the 1969 NFL/AFL draft. With the Bills, he received five consecutive Pro Bowl and first-team All-Pro selections from 1972 to 1976. He led the league in rushing yards four times, in rushing touchdowns twice, and in points scored in 1975. Simpson became the first NFL player to rush for more than 2,000 yards in a season—earning him 1973's NFL Most Valuable Player (MVP) award—and he is the only player to do so in a 14-game regular season. He also holds the record for the single-season yards-per-game average, at 143.1.

Murder trial of O. J. Simpson

Which prob. should the jury care about?

Pr (DNA match | innocent)

Or

Pr (guilty | DNA match)

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Murder trial of O. J. Simpson

Prosecutor's Bias

- DNA experts say

$$Pr(\text{DNA match} \mid \text{innocent}) = \frac{1}{170 \text{ Million}}$$

unrelated person

- Jury wants to know

$$1 - Pr(\text{guilty} \mid \text{DNA match})$$

to think they are equivalent is Prosecutor's Bias

- Bayes' Rule tells us

$$Pr(\text{guilty} \mid \text{DNA match}) \propto Pr(\text{DNA match} \mid \text{guilty}) \times Pr(\text{guilty})$$

Base rate

No fair trial will rely solely on DNA evidence

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Next Class

- Random variables
  - Discrete, continuous
  - PMF/PDF/CDF

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Thank you!